

CS587: Statistical and Mathematical Methods for Data Analysis

PROJECT PROPOSAL

Submitted By:

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1. PROJECT CATEGORY AND SPECIFIC IDEA

Project Category: Category V – Optimization, Priors, and Regularization

Specific Project: Feature Selection with L1 Regularization (Lasso)

Project Title: *Comparative Analysis of L1 and L2 Regularization: An Optimization Perspective on Automated Feature Selection in High-Dimensional Housing Data*

2. PRIMARY RESEARCH QUESTION AND OBJECTIVES

Research Question: How do L1 and L2 regularization differ in their ability to perform automatic feature selection while maintaining predictive performance, and what is the relationship between regularization strength and model sparsity?

Specific Objectives:

1. Formulate housing price prediction as a constrained optimization problem with different penalty functions (L1 vs L2 norms)
2. Implement and compare three approaches: (a) Ordinary Least Squares (OLS) – baseline, (b) Ridge Regression (L2) – shrinkage without sparsity, (c) Lasso Regression (L1) – shrinkage with automatic feature selection
3. Analyze how regularization affects model complexity, prediction accuracy, and feature selection effectiveness
4. Establish the connection between regularization and Bayesian priors ($L2 \leftrightarrow \text{Gaussian}$, $L1 \leftrightarrow \text{Laplacian}$)
5. Quantify the bias-variance tradeoff through cross-validation and determine optimal hyperparameter values

3. DATASET DESCRIPTION

Dataset: Ames Housing Dataset | **Source:** Kaggle | **Type:** Real-world

Characteristics: 1,460 residential properties in Ames, Iowa with 79 explanatory variables (continuous, categorical, and ordinal features) and target variable SalePrice ranging from \$34,900 to \$755,000.

Justification: (1) High-dimensional nature (79 features) ideal for demonstrating L1 feature selection capabilities, (2) Mix of relevant and irrelevant features allows clear demonstration of automatic

elimination, (3) Well-documented benchmark dataset ensuring reproducibility, (4) Real-world statistical properties (multicollinearity, varying feature importance) provide realistic optimization challenges, (5) Standard educational dataset allowing focus on methodology rather than data collection.

Preprocessing Plan: Handle missing values through median/mode imputation, encode categorical variables using one-hot encoding, standardize features for fair comparison, and split data 80/20 for training/testing with 5-fold cross-validation.

4. PRIMARY STATISTICAL AND MATHEMATICAL METHODS

A. Optimization Framework (CLO-3):

- Problem formulation: $J(\beta) = ||y - X\beta||_2^2 + \lambda R(\beta)$, where $R(\beta) = ||\beta||_2^2$ (Ridge) or $||\beta||_1$ (Lasso)
- Optimization algorithms: Closed-form for Ridge, coordinate descent for Lasso
- Hyperparameter optimization: Grid search with cross-validation over $\lambda \in [10^{-4}, 10^4]$

B. Bayesian Inference and MAP Estimation:

- Ridge penalty $||\beta||_2^2 \leftrightarrow$ Gaussian prior $p(\beta) \propto \exp(-||\beta||_2^2/2\sigma^2)$
- Lasso penalty $||\beta||_1 \leftrightarrow$ Laplacian prior $p(\beta) \propto \exp(-||\beta||_1/\tau)$
- Demonstrate regularized regression as MAP estimation: $p(\beta|y) \propto p(y|\beta) \times p(\beta)$

C. Linear Algebra and Multivariate Statistics:

- Matrix formulation $y = X\beta + \varepsilon$
- Coefficient shrinkage patterns and multicollinearity analysis

D. Statistical Estimation Theory:

- Bias-variance decomposition: optimal λ minimizes $\text{Bias}^2 + \text{Variance}$
- K-fold cross-validation (K=5) for unbiased performance estimation

E. Performance Evaluation:

- Prediction metrics: RMSE, R^2 , MAE
- Model complexity: sparsity (non-zero coefficients), L0/L1/L2 norms
- Feature selection quality: stability analysis across CV folds

Expected Deliverables: Comprehensive comparison of OLS/Ridge/Lasso, regularization path plots, cross-validation curves, feature importance rankings, sparsity analysis, quantitative evaluation (RMSE, R^2), visual demonstration of L1 vs L2 effects, Bayesian interpretation, and practical recommendations.